# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 1

**Full Syllabus (Chapters 1–10)** | Class 6 Maths | Maximum Marks: 50

### MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 21 | Each term is the sum of the two terms before it. |
| 2 | **(a)** 90 | At full hours the minute hand is at 12; each hour mark is $30^\circ$. |
| 3 | **(c)** 55 | Reverse and add. |
| 4 | **(b)** 4.5 | Divide by the value of one symbol. |
| 5 | **(b)** 3516 | Apply the divisibility rule of 4. |
| 6 | **(c)** 40 | Area $= \frac{1}{2} \times$ base $\times$ height. |
| 7 | **(b)** $\frac{5}{6}$ | Convert to a common denominator first. |
| 8 | **(b)** outside the circle | Compare 9 cm with the radius 5 cm. |
| 9 | **(b)** 2 | Along its two diagonals. |
| 10 | **(a)** $-5$ | Same signs: add and keep sign; different signs: subtract. |
| 11 | **(d)** 3 | Number of lines $= \frac{3\times2}{2}$. |
| 12 | **(d)** acute | Classify the turn. |
| 13 | **(a)** 5175 | Arrange digits in descending and ascending order and subtract. |
| 14 | **(b)** 8 | Divide the number by the scale. |
| 15 | **(d)** 8 and 15 | Co-prime numbers have no common factor other than 1. |
| 16 | **(a)** 24 | Half the product of the legs. |
| 17 | **(a)** $\frac{5}{12}$ | Common denominator first. |
| 18 | **(c)** 4 | Midpoint of AB. |
| 19 | **(a)** 72 | $360^\circ \div$ number of sides. |
| 20 | **(a)** $-7$ | Add the opposite. |
| 21 | **(a)** 25 | Two consecutive triangular numbers add up to a square number. |
| 22 | **(a)** 75 | Add the adjacent angles. |
| 23 | **(d)** 9 | Even: halve it. Odd: multiply by 3 and add 1. |
| 24 | **(a)** 20 | Subtract the smaller number from the larger. |
| 25 | **(c)** 11 and 13 | Twin primes are primes that differ by 2. |
| 26 | **(a)** 32 | Find the side first. |
| 27 | **(c)** $\frac{2}{3}$ | Compare using a common denominator. |
| 28 | **(c)** 5 | Diagonals of a rectangle are equal and bisect each other. |
| 29 | **(d)** the same letter | Vertical-axis symmetry keeps it the same. |
| 30 | **(b)** -6 | Right is positive, left is negative. |
| 31 | **(c)** 120 | Triangular number $T_{15} = \frac{15\times16}{2}$. |
| 32 | **(c)** 270 | Remaining turn $= 360^\circ -$ turn. |
| 33 | **(a)** 2 | Keep reversing and adding. |
| 34 | **(b)** 75 | Add all. |
| 35 | **(d)** 6 | Check each number. |
| 36 | **(a)** 9 | Breadth = area $\div$ length. |
| 37 | **(c)** $3\frac{5}{6}$ | Add the whole parts and fraction parts separately. |
| 38 | **(b)** 60 | Triangle OAB is isosceles with OA = OB. |
| 39 | **(d)** 4 | Order = $360 \div$ smallest angle. |
| 40 | **(a)** 8 | Count the integers strictly between. |
| 41 | **(b)** 256 | $16 \times 16$. |
| 42 | **(c)** 6 | Each pair of rays makes an angle. |

| | | |
|---|---|---|
| **43** | **(b)** 22 | Add all the digits. |
| **44** | **(c)** 2 | Count how many times each value appears. |
| **45** | **(c)** 5 | $72 = 2 \times 2 \times 2 \times 3 \times 3$ |
| **46** | **(a)** 1200 | Cost $=$ perimeter $\times$ rate. |
| **47** | **(a)** $\frac{2}{3}, \frac{3}{4}, \frac{5}{6}$ | Use a common denominator. |
| **48** | **(c)** 4 | Subtract the radii. |
| **49** | **(b)** 7 | Sides = lines of symmetry. |
| **50** | **(c)** $-9$ | Add the rise. |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

**Full Syllabus (Chapters 1–10)** | Class 6 Maths | Maximum Marks: 50

### MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 45 | Triangular numbers: term $n$ is $1+2+\cdots+n$. |
| 2 | **(b)** 145 | Inner and outer scales add up to $180^\circ$. |
| 3 | **(c)** 31 | Add the two cells below. |
| 4 | **(a)** 6 | Count the values bigger than the given number (not equal to it). |
| 5 | **(d)** 3 | The next un-crossed number after 2 is 3. |
| 6 | **(b)** 300 | Divide the floor area by the area of one tile. |
| 7 | **(c)** 3 | Reduce the fraction. |
| 8 | **(a)** equal and perpendicular bisectors of each other | Property |
| 9 | **(c)** Z | Z fits after a half turn but cannot be folded. |
| 10 | **(a)** 0 | $n+(-n)=0$. |
| 11 | **(d)** 216 | Cube numbers are $1^3, 2^3, 3^3, \ldots$ |
| 12 | **(b)** collinear points | Definition / property |
| 13 | **(a)** 19 | Middle row: $a+b$ and $b+c$; top $= a+2b+c$. |
| 14 | **(b)** 55 | Count the half symbol as half the value. |
| 15 | **(d)** 3 | Test 2, 3, 5, 7, ... |
| 16 | **(c)** 300 | 1 m = 100 cm. |
| 17 | **(a)** $\frac{3}{4}$ | Equal sharing: divide 3 by the number of children. |
| 18 | **(d)** 2 | Two pairs of equal opposite sides. |
| 19 | **(a)** 8 | Order $= 360^\circ \div$ angle. |
| 20 | **(d)** $50$ | Debt is negative. |
| 21 | **(d)** 34 | Each term is the sum of the two terms before it. |
| 22 | **(c)** 180 | At full hours the minute hand is at 12; each hour mark is $30^\circ$. |
| 23 | **(c)** 77 | Reverse and add. |
| 24 | **(b)** frequency | Definition |
| 25 | **(c)** 5128 | Apply the divisibility rule of 8. |
| 26 | **(b)** 42 | Area $= \frac{1}{2} \times$ base $\times$ height. |
| 27 | **(c)** $\frac{11}{12}$ | Convert to a common denominator first. |
| 28 | **(a)** 30 | $2(l+b)$. |
| 29 | **(d)** 0 | Symmetry property |
| 30 | **(a)** $-17$ | Same signs: add and keep sign; different signs: subtract. |
| 31 | **(d)** 6 | Number of lines $= \frac{4\times3}{2}$. |
| 32 | **(a)** right | Classify the turn. |
| 33 | **(b)** 7353 | Arrange digits in descending and ascending order and subtract. |
| 34 | **(b)** 3.5 | Divide by the value of one symbol. |
| 35 | **(a)** 14 and 25 | Co-prime numbers have no common factor other than 1. |
| 36 | **(b)** 30 | Half the product of the legs. |
| 37 | **(a)** $\frac{1}{3}$ | Common denominator first. |
| 38 | **(c)** outside the circle | Compare 8 cm with the radius 5 cm. |
| 39 | **(d)** 60 | $360^\circ \div$ number of sides. |
| 40 | **(b)** $5$ | Add the opposite. |
| 41 | **(c)** 36 | Two consecutive triangular numbers add up to a square number. |

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|---|---|---|
| **42** | **(a)** 80 | Add the adjacent angles. |
| **43** | **(d)** 34 | Even: halve it. Odd: multiply by 3 and add 1. |
| **44** | **(c)** 9 | Divide the number by the scale. |
| **45** | **(b)** 17 and 19 | Twin primes are primes that differ by 2. |
| **46** | **(c)** 44 | Find the side first. |
| **47** | **(a)** $\frac{7}{9}$ | Compare using a common denominator. |
| **48** | **(a)** 7 | Midpoint of AB. |
| **49** | **(a)** the same letter | Vertical-axis symmetry keeps it the same. |
| **50** | **(c)** 9 | Right is positive, left is negative. |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

**Full Syllabus (Chapters 1–10)** | Class 6 Maths | Maximum Marks: 50

### MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 210 | Triangular number $T_{20} = \frac{20\times21}{2}$. |
| 2 | **(b)** 225 | Remaining turn $= 360^\circ -$ turn. |
| 3 | **(a)** 3 | Keep reversing and adding. |
| 4 | **(d)** 20 | Subtract the smaller number from the larger. |
| 5 | **(a)** 7 | Check each number. |
| 6 | **(b)** 6 | Breadth = area ÷ length. |
| 7 | **(c)** $3\frac{3}{4}$ | Add the whole parts and fraction parts separately. |
| 8 | **(d)** 8 | Diagonals of a rectangle are equal and bisect each other. |
| 9 | **(b)** 5 | Order = $360 \div$ smallest angle. |
| 10 | **(c)** 9 | Count the integers strictly between. |
| 11 | **(d)** 289 | $17 \times 17$. |
| 12 | **(a)** 10 | Each pair of rays makes an angle. |
| 13 | **(b)** 35 | Add all the digits. |
| 14 | **(d)** 150 | Add all. |
| 15 | **(c)** 5 | $120 = 2 \times 2 \times 2 \times 3 \times 5$ |
| 16 | **(d)** 2400 | Cost = perimeter × rate. |
| 17 | **(c)** $\frac{3}{8}, \frac{5}{12}, \frac{1}{2}$ | Use a common denominator. |
| 18 | **(b)** 50 | Triangle OAB is isosceles with OA = OB. |
| 19 | **(c)** 9 | Sides = lines of symmetry. |
| 20 | **(b)** $-14$ | Add the rise. |
| 21 | **(b)** 78 | Triangular numbers: term $n$ is $1 + 2 + \cdots + n$. |
| 22 | **(a)** 125 | Inner and outer scales add up to $180^\circ$. |
| 23 | **(d)** 72 | Add the two cells below. |
| 24 | **(c)** 2 | Count how many times each value appears. |
| 25 | **(d)** 7 | Primes $p$ with $p \times p \le n$. |
| 26 | **(c)** 300 | Divide the floor area by the area of one tile. |
| 27 | **(a)** 5 | Reduce the fraction. |
| 28 | **(d)** 4 | Subtract the radii. |
| 29 | **(d)** H | H has two fold lines. |
| 30 | **(d)** positive | Distance is positive. |
| 31 | **(b)** 343 | Cube numbers are $1^3, 2^3, 3^3, \ldots$ |
| 32 | **(d)** initial point (vertex) | Definition / property |
| 33 | **(b)** 16 | Middle row: $a + b$ and $b + c$; top $= a + 2b + c$. |
| 34 | **(c)** 7 | Count the values bigger than the given number (not equal to it). |
| 35 | **(b)** 11 | Test 2, 3, 5, 7, ... |
| 36 | **(b)** 700 | 1 m = 100 cm. |
| 37 | **(a)** $\frac{3}{5}$ | Equal sharing: divide 3 by the number of children. |
| 38 | **(a)** are equal but not perpendicular | Property |
| 39 | **(b)** 6 | Order $= 360^\circ \div$ angle. |
| 40 | **(c)** $-25$ | Debt is negative. |
| 41 | **(b)** 55 | Each term is the sum of the two terms before it. |
| 42 | **(b)** 90 | At full hours the minute hand is at 12; each hour mark is $30^\circ$. |

| | | |
|---|---|---|
| **43** | **(b)** 99 | Reverse and add. |
| **44** | **(b)** 70 | Count the half symbol as half the value. |
| **45** | **(c)** 4218 | Apply the divisibility rule of 6. |
| **46** | **(b)** 63 | Area $= \frac{1}{2} \times$ base $\times$ height. |
| **47** | **(d)** $\frac{11}{12}$ | Convert to a common denominator first. |
| **48** | **(b)** 6 | It equals the side. |
| **49** | **(a)** 1 | Symmetry property |
| **50** | **(b)** $-7$ | Same signs: add and keep sign; different signs: subtract. |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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